Task 4 — Adverse Drug Reaction (ADR) Data Analysis

Virtual Works Lab | B.Pharmacy Internship Project

Project Objective

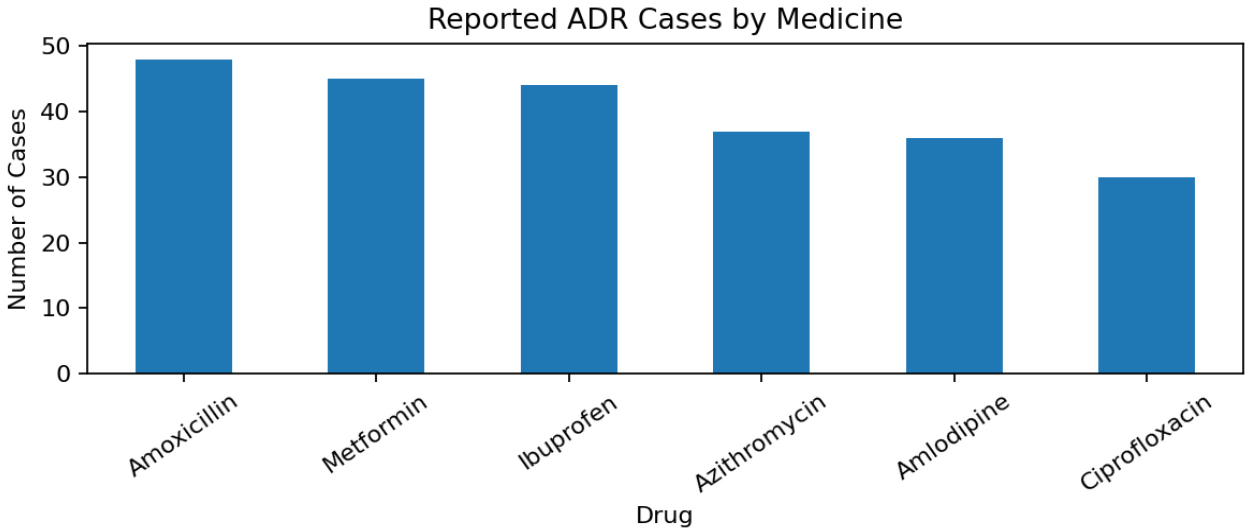
Analyze healthcare data containing patient symptoms, medication usage, and timelines to identify possible adverse drug reactions, correlate symptoms with drugs, identify higher-risk medicines, and evaluate possible causes.

Dataset

Synthetic educational ADR dataset containing 240 simulated reports with age, gender, medicine, symptom, onset time, severity, and outcome.

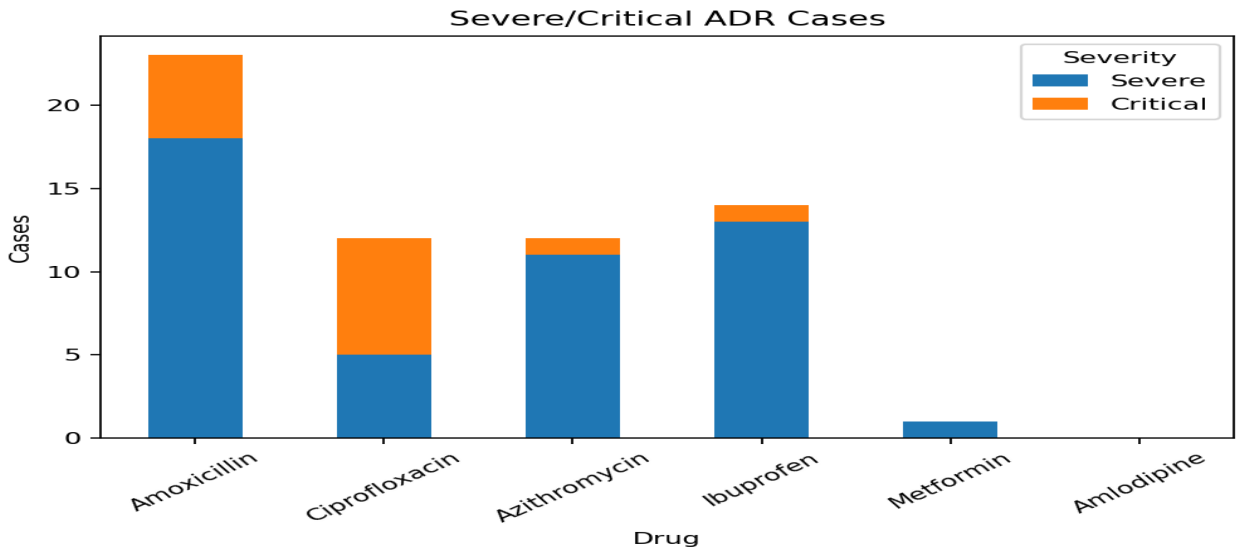
1. ADR Frequency by Medicine

Drug	ADR_Cases
Amoxicillin	48
Metformin	45
Ibuprofen	44
Azithromycin	37
Amlodipine	36
Ciprofloxacin	30



2. Severity and Higher-Risk Medicines

Mild	Moderate	Severe	Critical	Serious_%
8	17	18	5	47.92
5	13	5	7	40.0
8	17	11	1	32.43
15	15	13	1	31.82
32	12	1	0	2.22
18	18	0	0	0.0



3. Symptoms and Drug–Symptom Relationship

The most frequent symptoms were identified from the synthetic reports. A drug–symptom cross-tabulation was also calculated to identify patterns for further investigation.

Symptom	Cases
Nausea	54
Fatigue	49
Rash	45
Headache	34
Diarrhea	30
Dizziness	28

• Most frequently reported symptom: Nausea (54 cases).

• The serious-case percentage (Severe + Critical) is used as a simple screening indicator, not as a clinical risk estimate.

• Possible contributors to an ADR can include drug pharmacology, dose, patient characteristics, interactions, and underlying disease.

• Frequency or association alone does not prove that a medicine caused a symptom.

Drug–Symptom Cross-Tabulation

Drug	Diarrhea	Dizziness	Fatigue	Headache	Nausea	Rash
Amlodipine	4	2	9	8	8	5
Amoxicillin	6	6	5	8	13	10
Azithromycin	4	7	9	6	10	1
Ciprofloxacin	8	2	4	4	6	6
Ibuprofen	6	7	13	3	5	10
Metformin	2	4	9	5	12	13

Conclusion

The analysis demonstrates basic pharmacovigilance-style data exploration. In real pharmacovigilance, causality assessment requires clinical review and validated methods; this synthetic project is not a substitute for clinical assessment.